

Supplementary Information

S1 Current C protocol and selected actors

The C prior uses Qwen2.5-VL-7B-Instruct revision cc594898137f460bfe9f0759e9844b3ce807cfb5, P0 without a new system message, clean-plus-R1 image order, bfloat16, SDPA, deterministic generation and at most 500 new tokens. R1 rotates the RGB source clockwise once, limits the longest edge to 1024 pixels and applies readable row-major labels 0-63. A strict parser permits one format repair. The 211-row TRAIN/VALID prior is complete; TEST was not accessed.

Table S1. Current C actor selection.

Method	Seed	Selected update	VALID A (MPa)	Logical updates	Actual updates
C spatial feedback (main)	2026091301	250	45.847	1250	1250
C spatial open-loop	2026091303	1250	44.907	1250	1250
C mean feedback	2026091305	250	44.926	1250	1250

All five candidate checkpoints per C actor remain archived. Selection retained the earliest candidate unless a later domain-equal area improved by more than 1e-12. The six controls are frozen historical rows, while the three A policies are stored separately from the 650-row primary matrix.

S2 Statistical estimands

Same-cost endpoint losses first average Random repeat losses within each physical specimen. Pooled MAE and RMSE then aggregate 50 physical specimens. A and early A instead average repeats within specimen, specimens within domain and the six domains equally. Capture-group bootstrap intervals use 5000 paired draws within domain with seed 2026091401. These intervals are exploratory and conditional on checkpoint selection.

Timing uses $g_t = (1 - c_t/0.25)(e_{t-1} - e_t)$ in four right-closed completion intervals. Negative terms are retained and the machine-readable evidence verifies $A = e_0 - \sum_t g_t$. Equal-quality analysis searches the earliest supported crossing on the five-cap and event-union grids; it retains unreached targets, recrossings and negative or undefined savings.

S3 Complete same-cost results

Method	Cap	MAE	RMSE	R2	Mean actual cost	Actual range
Center- first	0.000	58.551	75.778	0.429	0.000	0.000000- 0.000000
Center- first	0.062	47.885	64.882	0.581	0.062	0.061399- 0.062407
Center- first	0.125	48.827	65.932	0.568	0.111	0.109491- 0.125000
Center- first	0.188	49.094	66.277	0.563	0.173	0.171900- 0.187121
Center- first	0.250	48.027	65.223	0.577	0.235	0.234676- 0.235577

Method	Cap	MAE	RMSE	R2	Mean actual cost	Actual range
Geometry-spread	0.000	58.551	75.778	0.429	0.000	0.000000-0.000000
Geometry-spread	0.062	50.221	67.804	0.543	0.062	0.061399-0.062130
Geometry-spread	0.125	46.189	62.421	0.612	0.125	0.123164-0.124629
Geometry-spread	0.188	46.555	62.347	0.613	0.187	0.186034-0.187407
Geometry-spread	0.250	46.910	62.278	0.614	0.250	0.249269-0.250000
Serpentine	0.000	58.551	75.778	0.429	0.000	0.000000-0.000000
Serpentine	0.062	49.731	64.721	0.583	0.062	0.062130-0.062407
Serpentine	0.125	48.773	64.778	0.583	0.125	0.124260-0.124629
Serpentine	0.188	48.343	63.931	0.593	0.187	0.186391-0.186851
Serpentine	0.250	48.013	63.041	0.605	0.249	0.248521-0.249258
Random	0.000	58.551	75.778	0.429	0.000	0.000000-0.000000
Random	0.062	53.156	69.115	0.525	0.054	0.046897-0.062500
Random	0.125	49.153	65.038	0.579	0.119	0.109307-0.125000
Random	0.188	48.108	63.412	0.600	0.187	0.171598-0.187500
Random	0.250	46.886	61.941	0.618	0.247	0.234492-0.250000
Learned-static	0.000	58.551	75.778	0.429	0.000	0.000000-0.000000
Learned-static	0.062	46.465	62.720	0.609	0.062	0.062130-0.062500
Learned-static	0.125	48.064	63.994	0.593	0.125	0.124260-0.125000
Learned-static	0.188	46.784	62.570	0.611	0.187	0.187130-0.187500
Learned-static	0.250	46.993	62.168	0.616	0.235	0.234674-0.249991
No-VLM spatial feedback	0.000	58.551	75.778	0.429	0.000	0.000000-0.000000
No-VLM spatial feedback	0.062	43.324	60.878	0.631	0.058	0.046897-0.062500
No-VLM spatial feedback	0.125	43.164	60.169	0.640	0.120	0.109120-0.124998
No-VLM spatial feedback	0.188	42.928	58.388	0.661	0.183	0.171415-0.187407

Method	Cap	MAE	RMSE	R2	Mean actual cost	Actual range
No-VLM spatial feedback	0.250	42.385	56.534	0.682	0.247	0.234492-0.250000
C mean feedback	0.000	58.551	75.778	0.429	0.000	0.000000-0.000000
C mean feedback	0.062	44.512	62.520	0.611	0.060	0.046789-0.062407
C mean feedback	0.125	45.494	62.573	0.611	0.123	0.109489-0.124998
C mean feedback	0.188	44.242	61.218	0.627	0.185	0.171598-0.187407
C mean feedback	0.250	44.066	60.593	0.635	0.248	0.234492-0.250000
C spatial feedback (main)	0.000	58.551	75.778	0.429	0.000	0.000000-0.000000
C spatial feedback (main)	0.062	46.987	63.404	0.600	0.060	0.046897-0.062407
C spatial feedback (main)	0.125	44.840	60.087	0.641	0.122	0.109120-0.124816
C spatial feedback (main)	0.188	44.270	59.570	0.647	0.185	0.171898-0.187407
C spatial feedback (main)	0.250	43.711	57.756	0.668	0.248	0.234492-0.249999
C spatial open-loop	0.000	58.551	75.778	0.429	0.000	0.000000-0.000000
C spatial open-loop	0.062	44.710	60.238	0.639	0.060	0.046713-0.062407
C spatial open-loop	0.125	44.720	59.192	0.651	0.122	0.109120-0.124814
C spatial open-loop	0.188	43.962	58.034	0.665	0.185	0.171898-0.187500
C spatial open-loop	0.250	43.917	58.523	0.659	0.249	0.234492-0.250000

S4 Historical A and current C

Method	Metric	Historical A	Current C	A - C	95% interval
C spatial feedback (main)	area_mpa	45.110	45.847	-0.736	[-2.946, 1.455]
C spatial feedback (main)	early_area_mpa	49.642	50.339	-0.696	[-4.500, 3.118]
C spatial feedback (main)	end-point_abs_error_mpa	44.286	43.711	0.575	[-1.035, 2.450]

Method	Metric	Historical A	Current C	A - C	95% interval
C spatial open-loop	area_mpa	46.501	44.907	1.594	[-0.500, 3.720]
C spatial open-loop	early_area_mpa	51.048	49.154	1.894	[-0.977, 5.090]
C spatial open-loop	end-point_abs_error_mpa	44.822	43.917	0.905	[-1.122, 2.839]
C mean feedback	area_mpa	45.263	44.926	0.337	[-2.208, 2.988]
C mean feedback	early_area_mpa	50.390	48.657	1.733	[-1.727, 5.652]
C mean feedback	end-point_abs_error_mpa	43.611	44.066	-0.455	[-3.146, 2.411]

Positive A-minus-C values favour current C. All domain-specific values, endpoint paired contrasts and target grids are retained in the accompanying CSV files.

S5 Current C case illustrations

The three cases were fixed before this retraining release. Every panel below was regenerated from the selected current C main-policy trajectory and current C prior. They are process illustrations rather than independent replications.

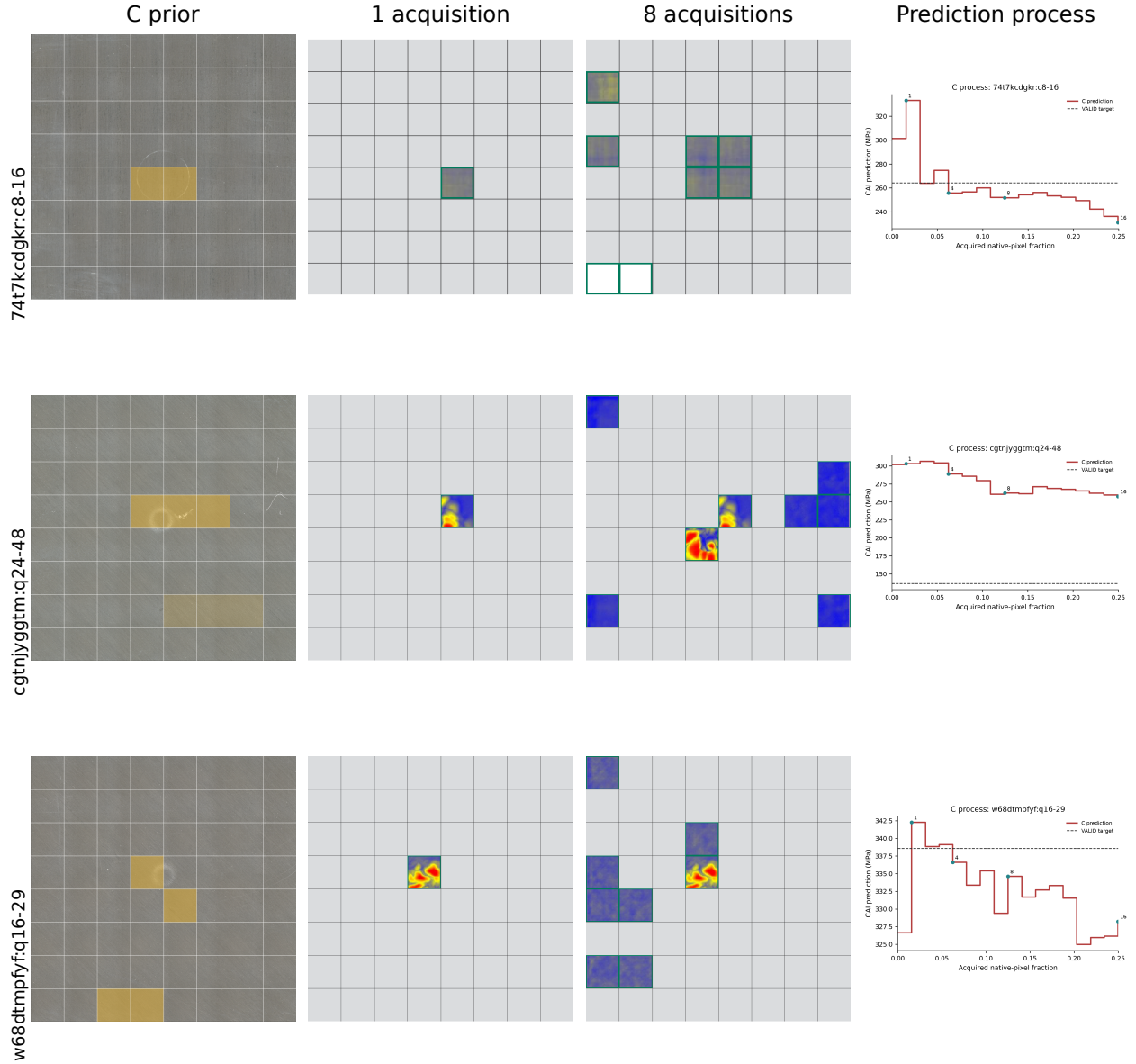


Figure S1: Current C candidate surfaces, acquired states and prediction processes for c8-16, q24-48 and q16-29.

S6 Source-data index

The tables directory contains the recomputed method summary, all five-cap metrics, domain metrics, paired contrasts, A/C version comparisons, event curves, timing contributions, quality targets, case states and full-input reference. The evidence manifest records their hashes. The figures directory contains vector PDF/SVG files and 300-dpi PNG previews, plus all new case PNGs.

References